Sunghyun **Park**

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"AI for Understanding the World, bridging science and technology to explore the unknown."

Education

Yonsei University — *Seoul, South Korea*
B.S. in Biochemistry and Computer Science (Double Major) — *Mar. 2020 – Present (Expected Aug. 2026)*

- GPA : 3.94/4.3 (Cumulative), 4.03/4.3 (CS Major)
- Oct. 2022- Jul. 2024 : Compulsory Military Service, Served as a sergent in R.O.K Air Force.

Research Interest

My research interest is in the learning dynamics of foundation models, with an emphasis on understanding how optimization shapes representation learning, generalization, and forgetting. I am particularly interested in developing mathematical and empirical frameworks for analyzing the dynamics of large language models during training. My current interests center on two directions: (1) **optimization dynamics** in post-training algorithms such as preference optimization and alignment, and (2) **machine unlearning**, focusing on how information is retained, forgotten, and redistributed throughout the optimization process.

Optimization | Machine Unlearning | LLM

Experience

Undergraduate Research, Mechanobiology Lab (Prof. Hyunkyu Choi) — *Seoul, South Korea*
Research Intern — *Dec. 2024 - Feb. 2025*

- Conducted high-resolution microscopy (TIRF, RICM) analysis of bioloical samples and gained hands-on experience with biophysical experimental techniques.
- Applied Unsupervised Machine Learning techniques to classify cells on their orientation.

Extracurricular Activity

PoolC (Yonsei Programming Club) — *Seoul, South Korea*
Member — *Mar. 2024 - PRESENT*

- Participated in AI-related paper studying groups.
- Organized seminars on the mathematical foundations of deep Learning - including information theory, optimization and Reinforcement learning.

Seoul-Learn (Mentoring Program) — *Seoul, South Korea*
Mentor — *Mar. 2024 - PRESENT*

- Guided middle and high school students in mathematics learning and academic planning.
- Provided guidance on choosing a major-overview of CS/AI and any science major curricula, prerequisites and study habits.

Biochemistry Student Academic Club — *Seoul, South Korea*
Member — *Mar. 2020 - Mar. 2022*

- Participated in collaborative study groups on core biochemistry subjects.
- Engaged in study of Organic Chemistry, focusing on reaction mechanisms through problem-solving sessions.
- Conducted group discussions on Immunology, deepening understanding of immune system functions and disease mechanisms.

Projects

Prompt Optimization with OPRO and Evolutionary algorithm — *Seoul, South Korea*
LLM Course project — *Jul. 2025*

- Implemented a prompt optimization loop combining **OPRO with an evolutionary algorithm** (selection, crossover, mutation).
- Used an LLM scorer to evaluate candidates and drive self-rewriting and selection.
- Evaluated on `GSM8K` with `Llama-3.1-8B-Instruct` and `Mistral-7B-Instruct`.

LLM Application | Prompt Engineering

Reducing WebOS Boot Latency through Build and Boot-Time Optimization *Seoul, South Korea*

YONSEI CAPSTONE DESIGN *Jun. 2025*

- Analyzed **function call traces** using `ftrace` to identify critical bottlenecks in WebOS boot process.
- Distinguished kernel-level and user-level execution delays, contributing to optimization of binary layout and page fault handling.

Operating System · Embedded System · Boot-Time Optimization

Force-Driven Mechano-Typing of B Cells: Multi-Channel Image Analysis of CD40-CD40L Interactions Using autoencoder *Seoul, South Korea*

UNDERGRADUATE RESEARCH INTERN *Dec. 2024 - Feb. 2025*

- Built a dataset with multi-threshold molecular tension probe sensors and microscopy for CD40-CD40L mechanics in B cells and automated single-cell segmentation.
- Compressed each cell image data into a low dimension and estimated local intrinsic dimension for each cell data.
- Visualized the global landscape with `UMAP` and conditional-wise trajectories.

AI For Science · Immunology · Computer Vision

Honors & Awards

DOMESTIC

2021 **Honors Award**, Yonsei University *Seoul, South Korea*

"Honors Award" is granted to students with outstanding academic achievement, typically ranking in the top 10%.

Relevant Coursework

CS/AI	Discrete Mathematics, Data Structures, Introduction of Computer science, Computer Architecture, Linear Algebra, **Machine Learning**, Algorithm Analysis, Computer Network, **Computer Vision**, Operating System, **Large Language Model**, Cloud Computing ,Introduction of Deep Learning, **Mathematics for Deep Learning**, **Distributed Learning and inference**, **Reinforcement Learning, Data privacy**
Biochem	Biochemistry, Organic Chemistry, Neuroscience, **Biophysical chemistry**, Biological Statistics, Stem Cell Biology

** Courses in **bold** indicate areas of particular interest.*

Skills

DevOps	AWS, Docker
Programming	Python, JAVA, C/C++, MATLAB, LaTeX
Framework	Pytorch, OpenCV, PyMOL
Languages	Korean (Native) , English (Intermediate)